

AngioSeal 碰到困難怎麼處理？

When AngioSeal Goes Wrong — Bail-out & Ultrasound Prevention

研討會報告

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關鍵回顧：Shariq 2023, J Clin Imaging Sci

為什麼要談這題

- AngioSeal (collagen-plug 型 VCD) 置放成功率 ~**98%**，但**失敗時併發症風險約 5 倍**
- 動脈阻塞 / 急性肢體缺血 (ALI) 雖罕見 (~**0.3%**)，卻可能**致殘、需手術**
- 本報告聚焦兩件事：
 - i. **Bail-out**：困難 / 併發症當下怎麼救
 - ii. **超音波預防**：怎麼從一開始就避免掉進坑

核心觀念：**同一個敵人貫穿全場——穿刺處的鈣化斑塊**（前壁或後壁）

AngioSeal 的三個元件 = 三個故障點

元件	正常功能	故障模式
錨片 (footplate/anchor)	動脈內壁定錨	栓塞 (脫落順血流) 、卡在鈣化斑無法回收 → 管腔內展開
膠原塞 (collagen plug)	動脈外止血	突入管腔 → 狹窄 / 阻塞 ; 感染巢
可吸收縫線 (suture)	夾合三明治	張力不足 → 錨片鬆脫漂浮

- 膠原塞約 **60–90 天**自行吸收 → 是「watchful waiting / 少放支架」的依據
- 機制核心：**後壁 / 前壁鈣化斑塊阻礙錨片回收**，footplate 在管腔內展開或脫落

誰容易失敗（風險因子）

- **血管條件**：總股動脈 (CFA) 嚴重鈣化 / 粥狀硬化、管徑小、周邊動脈疾病
- **穿刺**：低位（落 SFA/分叉）、antegrade、非 CFA
- **操作**：大號鞘、重複部署、長手術時間、抗凝
- **族群**：肥胖、糖尿病、高齡

Abando 法則：血管 $< 5\text{ mm}$ 或穿刺處附近狹窄 $\geq 40\%$ → **不要用 AngioSeal**
(感染風險 0.2–0.6% ; ASCD 併發症 up to ~2.5% , 多在鈣化血管)

流行病學與時間

Epidemiology & Timing

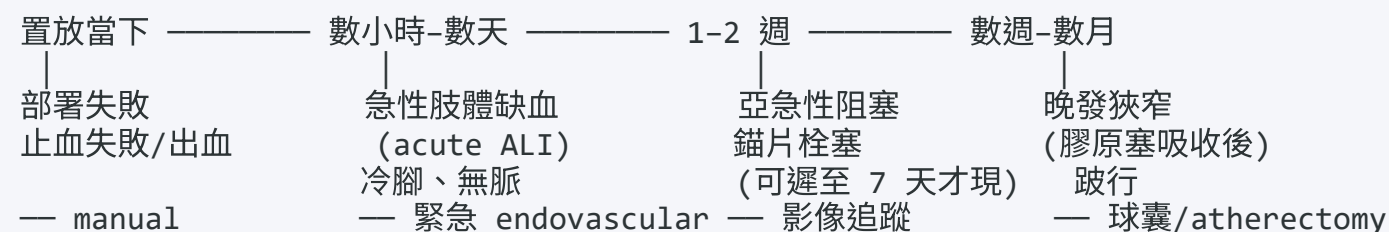
失敗率與後果 (Bangalore 2009, n=9823)

- VCD 失敗率 **2.7%** (診斷 2.3% vs PCI 3.0%)
- **失敗 → 血管併發症顯著上升：**

結果	失敗組	成功組
任何血管併發症	6.7%	1.4%
主要併發症	1.9%	0.6%
肢體缺血	0.4%	0.1%
假性動脈瘤	0.7%	0.1%

配對後任何併發症 OR **5.63**。失敗的 VCD 比單純手壓更危險 → 這些病人必須密切監測。
限制：僅院內結果，**晚發併發症被低估** (見下頁時間軸)

併發症的「時間軸」 (Timing)



- ALI 發作分布 (Haberman, n=40) : 急性 (同次住院) 48% / 亞急性 <2 週 30% / 慢性 >2 週 22%
- 中位診斷時間 13 天 ; 跛行 92% 、 ALI 65% ; 阻塞位置 CFA 88% / SFA 12%
- 錨片栓塞可遲至 7 天才以 ALI 表現 (Shariq Case 1)

併發症全覽 (Complications)

併發症	機制	處理主軸
急性肢體缺血 / 動脈阻塞	膠原塞突入 + 鈣化斑	endovascular (atherectomy + balloon) / 手術
錨片栓塞 (footplate)	脫落入 SFA/髖動脈	取出 (forceps) / 手術
晚發狹窄	局部纖維化	球囊 / atherectomy
假性動脈瘤	止血不全	超音波導引 thrombin/fibrin
出血 / 腹膜後出血	未達止血	壓迫、影像、輸血、介入
感染	膠原塞異物巢	抗生素 + 外科清創/切除
部署失敗	錨片未坐穩	勿強拉 → manual / 補救裝置

困難處理流程

The Bail-out Algorithm

困難處理流程圖 (Decision Tree)



兩條主軸：動脈阻塞走 endovascular (atherectomy 為主力) ；假性動脈瘤走超音波導引注射。
外科保留給：endovascular 失敗、巨大/感染、錨片無法取出。

Bail-out ①：部署當下的「旋轉手法」

情境： 部署時 footplate 卡在鈣化斑、阻力大、止血失敗 (Shariq Fig 6)

- **超音波**確認 footplate 是否卡在前/後壁鈣化斑
- 手法：**將裝置旋轉 180°** → 緩緩回收，直到 footplate **越過斑塊遠端** → 再轉回中位
- 仍不行 → **不要硬拉** (會把錨片拉斷/拉進管腔) → 改 **manual compression** 或補救裝置

預防 > 補救：**部署前用超音波看清楚斑塊位置**，是這一步成敗關鍵

Bail-out ②：急性缺血 / 阻塞 — Endovascular 主力

Haberman 2024 (n=40 , ALI 0.32%) — 目前最佳實務 ★

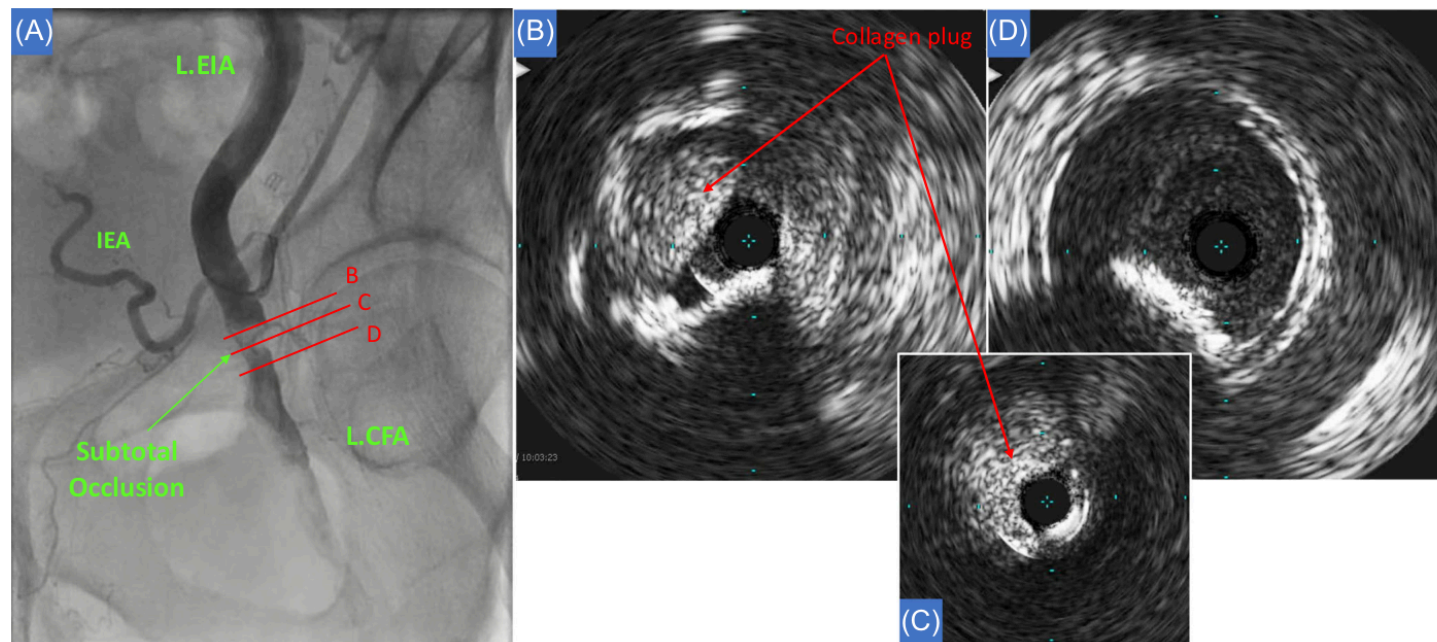
- 路徑：對側 crossover (100%) + 遠端栓塞保護 EPD 48%
- Directional atherectomy 88% (HawkOne/TurboHawk) 清除膠原塞 + balloon 100% ± cutting balloon 13% ± stent 18% (IVUS 43%)
- 結果：100% 成功、0 轉手術；中位追蹤 244 天 0 再介入

Dong 2017 (n=32 , ALI 0.29%) — balloon-first

- Balloon 58% / +溶栓(urokinase) 29% / +stent 13%；成功 96.9%；狹窄 97.5%→23.9%；再狹窄 9.7%

支架要克制：CFA 過髌關節有 fracture 風險；膠原塞會自溶 → 多數不需 stent。Cutting balloon 較適合縫線型 VCD 阻塞。

影像 | 阻塞的真面目：IVUS 上的膠原塞



血管攝影見 collagen plug remnant ; IVUS 顯示**膠原塞幾乎堵住管腔 + 鈣化結節**。來源：Haberman 2024, *Catheter Cardiovasc Interv* (教學用途)

影像 | Endovascular bail-out (atherectomy)

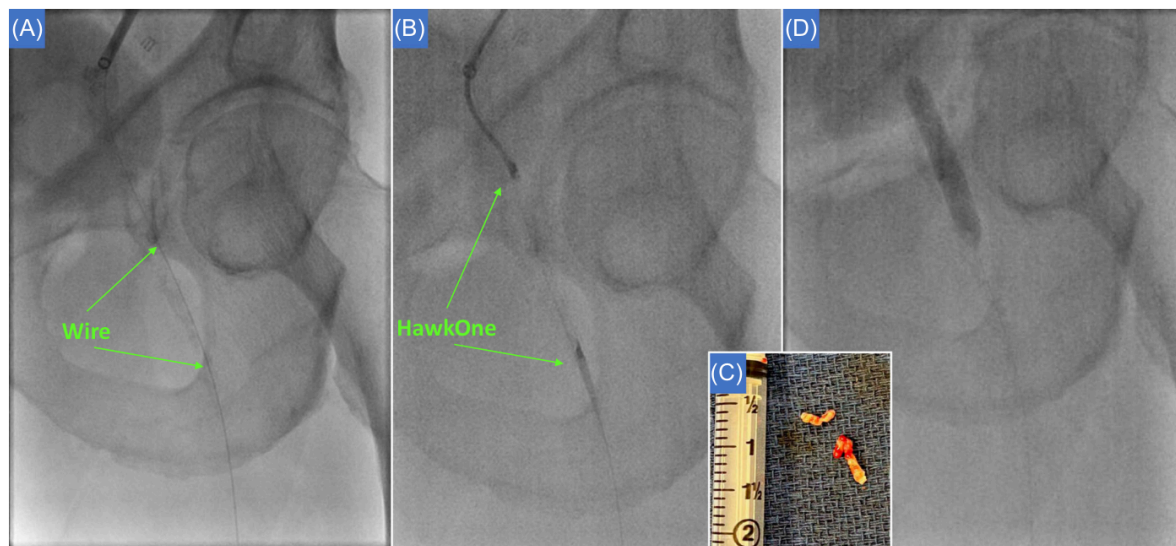


FIGURE 2 Percutaneous endovascular intervention. (A) The lesion is crossed with guidewire antegradely. (B) Directional atherectomy with HawkOne catheter was used to debulk the lesion. (C) Plug particles extracted from the HawkOne collection chamber. (D) The lesion is inflated with 1:1 balloon to vessel balloon. [Color figure can be viewed at wileyonlinelibrary.com]

(A) 導絲通過 (B) HawkOne directional atherectomy 清除膠原塞 (C) 取出的 plug 碎屑 (D) 球囊擴張。來源：Haberman 2024, *Catheter Cardiovasc Interv* (教學用途)

影像 | 急性肢體缺血：CTA 與 DSA (治療前後)

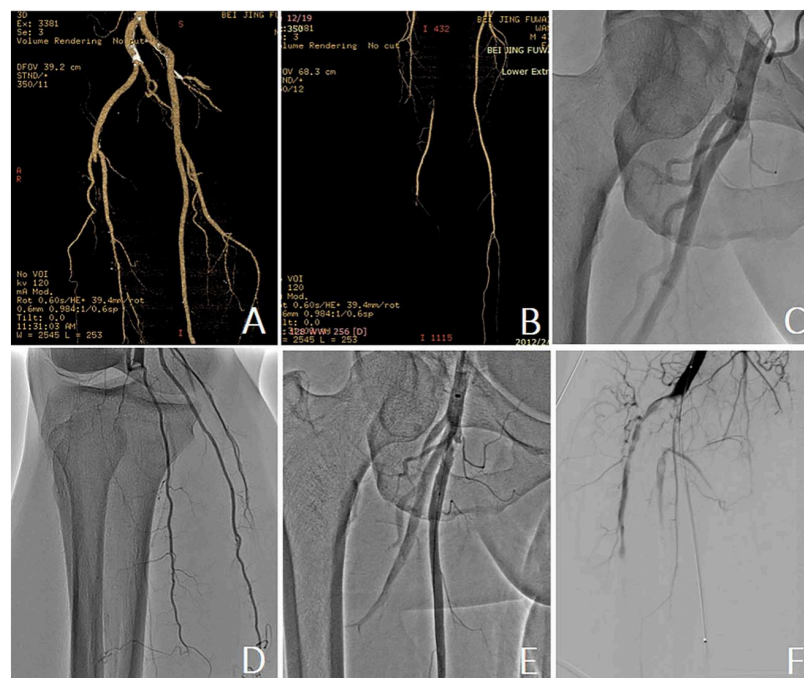


Fig. 4. Representative angiograms of a patient with acute limb ischemia. (A) 3D-CTA before treatment, (B) 3D-CTA after treatment, (C) DSA before treatment, (D) DSA after treatment, (E) DSA before treatment, (F) DSA after treatment.

3D-CTA 見 SFA 阻塞 + 膝下血栓；DSA 球囊擴張前後（狹窄 97.5%→23.9%）。來源：Dong 2017, *Catheter Cardiovasc Interv* (教學用途)

Bail-out ③：外科與錨片取出

- **外科** = endovascular 失敗、巨大/感染、錨片無法取出時的後盾
 - CFA endarterectomy + **footplate 取出** (Shariq Case 3：取出完整裝置)
 - 導線無法通過 → **ilio-popliteal bypass** (Dong 1 例)
- **錨片栓塞 / 漂浮 (floating Angio-Seal)**
 - 外科：縱向動脈切開取出 (Dettmers)
 - Endovascular：**Alligator Tooth Retrieval Forceps** 取出 (文獻替代)

教學點：**裝置不是放完就沒事**——Dettmers 個案是出院前理學檢查聽到 **CFA 收縮期雜音** 才發現！

Bail-out ④：假性動脈瘤 — 超音波導引注射（首選）

超音波導引 thrombin 注射 (UGTI) — 現代標準，成功率高

- 成功率 **85–100%** (Kurzawski 首針 85%、重複後 100% ; Ozawa 100% ; pooled ~97.5%) ; 再發 ~3.3%、併發症 1.3%
- **技術**：21G 針、bovine thrombin 1000 U/mL ; 先注 **0.2 mL (200 U)**，看 color flow，殘流再加 0.1–0.2 mL ; **終點 = 囊內血流消失 + 遠端動脈通暢**
- **不需停抗凝**也有效；務必全程看到針尖（避免打進動脈）

超音波導引 fibrin glue (Gummerer)：成功 87.3%、轉手術 12.7%、**遠端栓塞 4%**

影像 | 假性動脈瘤的超音波 (yin-yang)

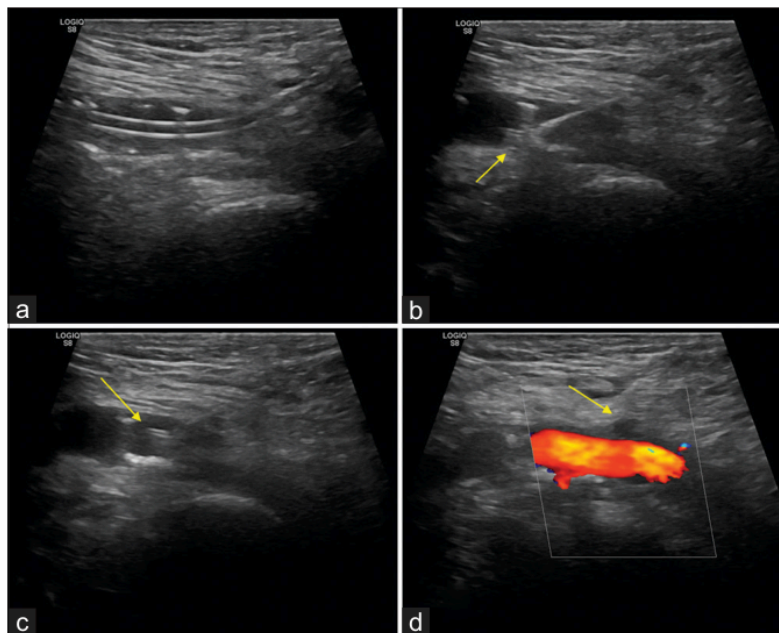


Figure 6: A 76-year-old man who underwent left leg angioplasty –

AngioSeal 後總股動脈假性動脈瘤；color-Doppler 見典型 yin-yang 血流。來源：Shariq 2023, *J Clin Imaging Sci* (open access)

影像 | 超音波導引 thrombin 注射

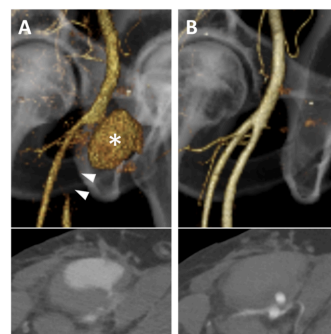


Fig. 1 (A) Digitally recreated three-dimensional (3D) image of computed tomography angiography shows the pseudoaneurysm of the common femoral artery (asterisk), complicated with severe stenosis of the superficial femoral artery due to compression by the pseudoaneurysm (arrowheads). (B) This is a 3D image 5 days after the ultrasound-guided thrombin injection, showing disappearance of the pseudoaneurysm and resolution of the external compression to the superficial femoral artery.

the PSA was detected during this 1-month

Case 2 (Table 1, Patient No. 6)

An 81-year-old male admitted to our lifestyle-limiting claudication of the right I class 2). He was a current smoker and hypertension and peripheral artery disease EVT on his left leg 9 months prior to admission taking aspirin. In this admission, he underwent stenting in the right external iliac artery at balloon angioplasty in the right SFA via sheath placed into the left CFA.

After removing the sheath, manual compression was applied at the puncture site. However, 30 minutes of compression failed to achieve hemostasis from the puncture site on the skin was reduced the compression. Therefore, the decision was made to proceed with UGTI, and manual compression was converted to compression using the ultrasound. A small extravascular space with arterial flow was seen in color flow image. A 21-gauge needle was then placed into the space under ultrasound guidance. Once blood flowed from the needle, 0.4 ml (400 IU) of thrombin solution was injected, resulting in immediate thrombosis of the extravascular space (Fig. 2B). The postoperative course was unremarkable with no recurrence of rebleeding events, and he was discharged 1 day after the procedure.

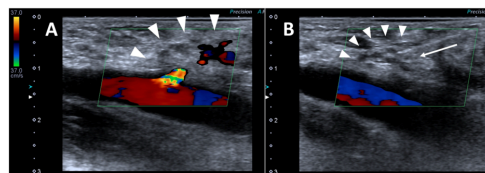


Fig. 2 (A) A small extravascular space with arterial flow via the arterial defect was seen in the

3D-CTA 假性動脈瘤 (星號) + 注射 5 天後消退 ; color-Doppler 見頸部血流 , 21G 針注入 0.4 mL (400 IU) thrombin 。 來源 : Ozawa 2022, *Ann Vasc Dis* (open access)

假性動脈瘤：選擇與紅線 (Cochrane + 共識)

- **Cochrane (Tisi 2013)**：壓迫為一線；**超音波導引壓迫並不優於盲壓**；壓迫失敗再用 thrombin。各別 RCT 偏好 thrombin，但**統合後未達顯著**（樣本小）
- **務實順序**：小（<2–3 cm）/無抗凝 → 觀察；其餘 → **UGTI**（最高成功率）
- **避免經皮注射、直接外科**：**neck <1 cm 或 defect >5 mm**、感染、皮膚壞死、破裂、巨大、AV fistula、肢體威脅

注意：Cochrane 指出**沒有任何 RCT 納入手術組**——「外科為金標準」缺乏 RCT 實證

超音波預防

Ultrasound Prevention — 從源頭避免困難

預防 ①：超音波導引「穿刺」

Foerschner 2022 (n=479 , 導管電燒) ★

- 總血管併發症 **6.3% vs 10.7%** ; 血腫 **>5 cm** 2.5% vs 6.3% 、 假性動脈瘤 0.6% vs 3.8%
- **肥胖 (BMI>30)** 獲益最大 (OR 0.138) ; ACT 劑量無影響 → **穿刺品質 > 抗凝控制**

Skalidis 2025 (n=231 , 周邊介入 , 全程 US 穿刺)

- VCD 併發症低於 manual compression (**MC vs VCD OR 2.41**) ; 穿刺處鈣化 OR **2.74** ; collagen-plug 裝置失敗 **0%**

技術：分辨動靜脈、在**分叉上方**穿刺、避開鈣化 / 後壁、即時看針尖

預防 ②：超音波導引「部署 / 止血」

US-MANTA (Miyashita 2022, n=1150) — 大號 VCD ★

- 超音波導引使血管併發症 12.5%→6.8% ($p=0.001$)、裝置失敗 7.5%→3.9% ($p=0.012$) ; US-guided OR 0.56 (0.36–0.88)
- 用長軸掃描定位 toggle、確認在 CFA 內、與前壁平行 $\geq 45^\circ$ 再釋放
- US 也讓裝置失敗能當下立刻辨識

US-guided 縫線止血 (Kwak) — 「heart-shape sign」

- 即時看 knot-trimmer 抵到前壁再剪線；止血時間 256 vs 317 秒、需補手壓 7/54 vs 15/50

超音波貫穿三階段：穿刺 → 部署確認 → 止血確認

影像 | 超音波導引止血：heart-shape sign

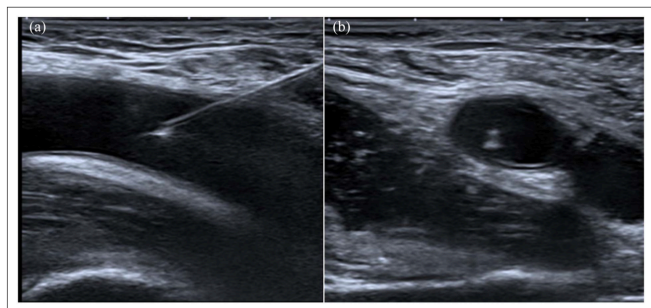
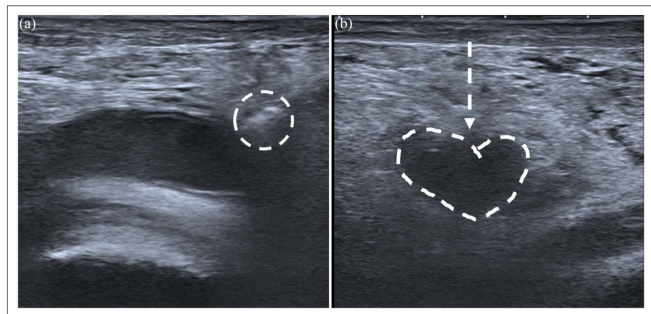


Figure 1. (a) Axial and (b) longitudinal view of US-guided common femoral artery puncture. In the axial view, it can be seen that the 12 o'clock direction of the common femoral artery is accurately punctured, and in the longitudinal view, it can be seen that there are no obstacles such as lymph nodes in the access route.



上：US 導引總股動脈穿刺（軸切 + 長軸）。下：knot-trimmer 抵前壁時動脈被壓成「心形」（b 箭頭） = 止血到位。來源：Kwak 2025, *J Vasc Access*（教學用途）

殘存的敵人：前壁鈣化 (The Residual Enemy)

- 超音波**降低**但**無法完全克服**前壁鈣化造成的失敗
- **兩個世代一致**：嚴重鈣化時 US 的保護效益**消失** (Miyashita n=1150 **p-interaction 0.048** ; Skolidis 鈣化 **OR 2.74**)
- 對策：
 - **長軸掃描**標定鈣化、避開前壁/側壁/分叉
 - 鈣化嚴重 → **改策略** (manual compression、改入路、或外科切開縫合)
 - 回到 **Abando 法則**：小血管/狹窄就**別硬上 AngioSeal**
- **永遠保留對側入路**作為 bail-out

反向思考：把問題變工具 — Transaneurysmal AngioSeal

Auer 2023 (n=14 , 多為 TAVI 後 CFA 假性動脈瘤)

- 對保守/壓迫/thrombin 失敗的假性動脈瘤，用 AngioSeal 經瘤頸封閉
- 關鍵手感：locator 側孔出血由噴射 → 涓滴 (spurt → trickle) = 針尖已在瘤頸；再進 1–2 cm 入動脈、釋放錨片、把膠原塞拉進瘤頸
- 技術成功 100%、零併發症、無需手術；隔日超音波確認囊內無血流、遠端正常

AngioSeal 不只是「製造困難」的裝置，必要時也能反過來成為 bail-out 工具

影像 | Transaneurysmal AngioSeal 技術

Fig. 1 Well-perfused PSA of the CFA in a 91-year-old patient (Patient No. 3) after TAVI. **A** Sonography showing persistent perfusion after single thrombin injection. **B** Puncture of the PSA close to its neck and DSA via the connecting catheter to visualize the patient's vascular anatomy. **C** Correct probing and positioning of the microwire in the pelvic axis. **D** Insertion of a 5 F sheath after changing to a 0035" wire and DSA showing only residual perfusion of the PSA, confirming correct position. **E** and **F** Sonography on the next day shows nonperfused PSA the next day

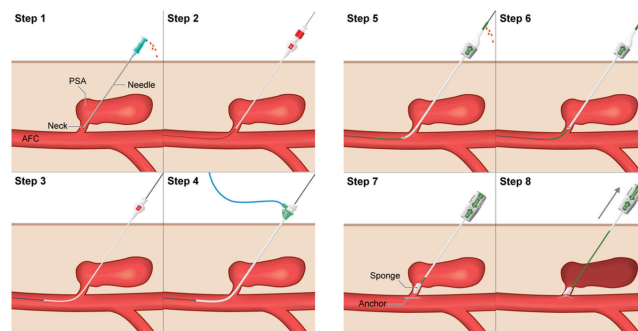
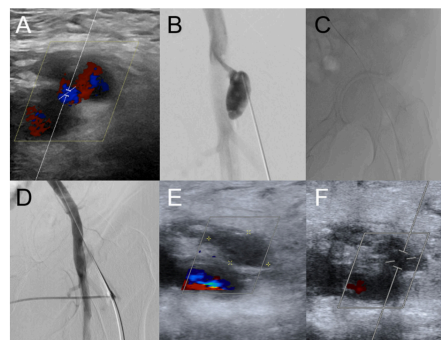


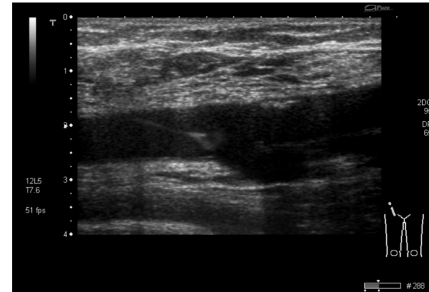
Fig. 2 Photos and schematic illustration of the procedure steps

Fig. 3 Photos and schematic illustration of the procedure steps

上：US/DSA 經瘤頸定位與封閉 (A-F)。下：Step 1-8 示意——穿刺瘤頸 → 導絲 → 鞘 → 釋放錨片 → 膠原塞拉入瘤頸封閉。來源：Auer 2023, *Cardiovasc Intervent Radiol* (教學用途)

影像 | 漂浮的 Angio-Seal (floating anchor)

ptoms or complications on the second postoperative day. Patient consent was obtained for publishing thi



Ultrasound of the common femoral artery, showing the luxated anchor of the Angio-SealTM floating in the lumen, tore from the vessel wall.



上：US 見管腔內漂浮的 luxated anchor (懸在縫線上) 。下：手術取出的完整裝置。來源：Dettmers 2016, *Open Cardiovasc Med J* (open access)

影像區（你自己的案例：請插入去識別化影像）

建議放入的影像（對照文獻所示）：

1. **超音波**：CFA 前/後壁鈣化斑；管腔內**echogenic footplate**；「漂浮錨片」；color-Doppler 假性動脈瘤 yin-yang
2. **IVUS**：膠原塞幾乎堵住管腔 + 鈣化結節（Haberman Fig 1）
3. **DSA**：CFA 次全阻塞 pre → atherectomy/balloon post；錨片栓塞至膈動脈
4. **CTA**：focal CFA 阻塞、below-knee thrombosis
5. **你自己的案例**：困難部署、bail-out 過程、術後通暢的血管攝影 / 超音波

（依你習慣：插入自己手術的去識別化 angiograph / US；本頁為版位占位）

代表性案例 (Illustrative Cases)

- **Case A (Shariq C1)** : 72F , DM/吸菸 , footplate 栓塞至髖動脈支架 , 7 天後才以 ALI 表現 → 溶栓 + **Supera 支架** , 6 月通暢
- **Case B (Shariq C3)** : 63F , 腦血管攝影後**急性冷腳**、CFA 阻塞 → **外科 endarterectomy + 取出 footplate**
- **Case C (Dettmers)** : 63M , **完全無症狀** , 出院前理學檢查聞 **CFA 收縮期雜音** → 超音波見漂浮錨片 → 擇期手術取出
- **Case D (Auer)** : TAVI 後假性動脈瘤 , 壓迫/thrombin 失敗 → **Transaneurysmal AngioSeal** 封閉成功

教學：出院前聽診/觸診穿刺處；早期辨識 = 保肢關鍵

Take-home Algorithm

1. **預防**：Abando 法則篩選；**超音波導引穿刺**（分叉上、避鈣化）；部署前看清斑塊
2. **部署困難**：超音波看 footplate → **旋轉 180°** 解卡 → 不行就 manual，**別硬拉**
3. **急性缺血/阻塞**：對側 crossover + EPD + **directional atherectomy + balloon**（Haberman）；支架克制；失敗→外科
4. **假性動脈瘤**：**超音波導引 thrombin**（85–100%）；neck<1cm/感染/巨大→外科
5. **錨片栓塞**：forceps 取出 / 外科
6. **永遠保留對側入路**；出院前檢查穿刺處

一句話：**鈣化是主因、超音波是預防、atherectomy 與 thrombin 是兩大 bail-out 主力。**

參考文獻 (已下載全文 · PubMed 查證)

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謝謝聆聽

Q & A

一句總結：鈣化是主因；超音波從穿刺到止血全程預防；
directional atherectomy 與**超音波導引 thrombin** 是阻塞與假性動脈瘤的兩大 bail-out 主力。

本投影片為研討會教學整理，供臨床討論，非臨床指引或正式醫囑。文獻全文已下載、PubMed 查證。